

Arcot Yashwanth

AI Engineer | Generative AI Engineer

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GitHub | LinkedIn

Summary

AI Engineer with 1+ year of experience building production-grade Generative AI applications, Retrieval-Augmented Generation (RAG) systems, and enterprise AI solutions using FastAPI, MongoDB Atlas Vector Search, LLMs, and AWS. Experienced in developing scalable AI backend services, document intelligence pipelines, and conversational AI applications.

Skills

Programming

Python, JavaScript, SQL, Bash

Generative AI

Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Tool Calling, Conversational Memory, LLMOps

Large Language Models

OpenAI, Gemini, Whisper, XTTS, spaCy, AllenNLP

Machine Learning

PyTorch, TensorFlow, Scikit-learn, XGBoost, SHAP

Backend

FastAPI, Django, Flask, REST APIs, WebSockets

Databases / Vector Search

MongoDB, PostgreSQL, MySQL, MongoDB Atlas Vector Search, FAISS, Pinecone

Cloud / DevOps

Docker, AWS (S3, Bedrock, SageMaker), GitHub Actions, n8n

Experience

AI Developer — Dealdox Software Pvt Ltd, Bangalore

May 2025 – Present

- Architected and deployed production-grade Retrieval-Augmented Generation (RAG) infrastructure using FastAPI and MongoDB Atlas Vector Search, enabling low-latency semantic retrieval across enterprise knowledge bases.
- Implemented conversational memory and coreference resolution pipelines using spaCy and AllenNLP, improving multi-turn contextual understanding in enterprise AI assistants.
- Built real-time embedding synchronization pipelines using MongoDB Change Streams, ensuring continuously updated enterprise knowledge bases without manual re-indexing.
- Built multilingual voice AI pipelines using Whisper, XTTS, and Gemini for transcription, translation, and voice synthesis.
- Built AI-powered document intelligence pipelines for CAD vector PDFs, reducing manual engineering document processing effort by **40%**.
- Automated AI document parsing workflows processing over **1,000+ engineering documents** with scalable extraction and indexing pipelines.
- Developed machine learning models on **60k+ CPQ records** using regression and classification algorithms with SHAP-based explainability.

Projects

PDF-based RAG AI Assistant

FastAPI, MongoDB Atlas Vector Search, OpenAI Embeddings, Railway

- Built an end-to-end RAG chatbot supporting PDF ingestion, semantic search, and conversational AI.
- Developed REST APIs for document ingestion, embedding generation, and vector search.
- GitHub | Live Demo

Web Host Manager Dashboard

React, Node.js, Django, MongoDB, REST APIs

- Built a scalable hosting management platform with REST APIs, RBAC, MongoDB, and responsive dashboards.
- GitHub | Live Demo

Multilingual AI Voice Translator

Whisper, XTTS, Python, Gemini

- Developed multilingual voice AI pipelines using Whisper, XTTS, and Gemini for speech transcription, translation, and neural voice synthesis.
- Developed scalable transcription and neural voice synthesis pipelines.

Education

B.Tech in Artificial Intelligence and Data Science

2021–2025

Nedurumalli Bala Krishna Reddy Institute of Technology